Table 5.1 Major Parts of an IC Engine and Its Material

Part	Material
Cylinder Block	Grey cast iron or Aluminum alloy
Cylinder Liner	Steel alloy containing Nickel, Manganese, Chromium, Silicon
Cylinder Head	Grey cast iron or Aluminum alloy
Crank Case	Grey cast iron or Aluminum alloy
Oil Sump	Pressed steel
Gasket	Alloy Steel containing copper, asbestos
Piston	Aluminum alloy
Piston Pin	Special steel
Piston Ring	Cast iron
Connecting Rod	Forged Alloy steel
Crank Shaft	Heat treated alloy steel
Vibration Damper	Aluminum alloy or wrought iron
Timing Gear, Timing Chain	Special alloy steel
Cam Shaft	Heat treated alloy steel
Main Bearing	Steel containing phosphor cronze, lead, bronze, tin, antimony and aluminum
Valve	Alloy steel containing silicon, chromium, nickel
Manifold	Cast iron or Aluminum
Flywheel	Pressed steel / cast iron
Silencer	Cast iron or hardened steel

5.3.1 Cylinder Block

The cylinder block is the main supporting structure for the various components. Cylinder block will have one or more cylinders. For multicylinder engine, the cylinders are cast as a single unit, called cylinder block. The cylinder block inner surface is machined and finished accurately for the piston to reciprocate up and down. The cylinder head is mounted on the top of the cylinder block. Cylinder head gasket is placed between the cylinder block and cylinder head. The cylinder head and cylinder block are provided with water jackets or with cooling fins. The crankshaft is mounted on bottom of the cylinder block with the help of bearings. The bottom portion of the cylinder block is called crankcase. A sump for lubricating oil is fastened to the bottom of the crankcase.

Cylinder blocks are two types

1. **Split block:** Crank case and cylinder block are manufactured as a separate unit and
2. **Mono block:** Crank case and cylinder block are manufactured as a single unit

The mono block and split block are shown in Fig 5.3.1.

5.3.2 Cylinder Liner

The cylinder will wear with use, and therefore there may be reduction in

AM_Unit_05.indd 80 12/20/2019 5:26:30 PM